(e) (iii) Gradient **1**

Ignore any units given with the value.

Hypotenuse of Δ must be > half the length of line drawn.

Check the read-offs. Work to half a small square. $A\Delta/A\Delta$ gets zero.

Values taken from the table that lie on the line to within half a small square are acceptable.

y-intercept **1**

(f) k = gradient, W = y-intercept **1**

Graph of I vs F will not score this mark unless analysis is consistent.

Unit of k and unit of W *(i.e.* N A^{-1} and N respectively) **1**

SF in k **1**

Allow 2 of 3 sf only

(g) Deducting the weight of the magnet from F will give a straight line passing through the origin and therefore force of attraction is proportional to current. **2**

If the weight of the magnet is not taken into consideration, then score zero.

Statement that force of attraction is not directly proportional to current scores zero.

(h) (i) Value of I when F = 10 N **1**

Working must be checked.

(ii) Overheating problems with the coil when F = 10 N **1**

Do not allow answers such as 'large voltage cannot be obtained from the psu'

20 marks in total

Question 2

A1 Procedure OK (i.e. measure P and f; change P and repeat). **1**

This mark can be scored even if the method is unworkable.

A2 Diagram of workable arrangement **2**

Source of sound + pump (1 mark); microphone + CRO (1 mark)

Allow ½ if all apparatus is inside the bell jar.

Allow ½ if the container is open.

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A3 Measurement of P **1**

(e.g. Bourdon gauge/pressure gauge/manometer/barometer). Must be shown correctly on the diagram.

B1 Correct measurements taken to find frequency using CRO **1**

Length of trace on screen + timebase setting

B2 Use of measurements to calculate frequency, or $f = 1/T$ **1**

B3 Maintain constant temperature whilst pressure is reduced **1**

OR maintain constant frequency as pressure is reduced

OR close tap before taking readings

C Safety precaution **1**

Safety screens/goggles

D1/2 Any further good design features **2**

Some of these might be:

Difficulty with detecting sounds of low intensity at low pressures

Use a signal generator connected to speaker

Vacuum grease the wires to the speaker

Allow time between readings for apparatus to warm up/cool down

Monitor temperature with thermometer during experiment

Avoid unwanted sounds/use soundproof room

Source of sound and microphone both inside the chamber

Increase P as well as decrease P to give wide spread of readings

10 marks in total

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